

WESTMINSTER
International University in Tashkent

System Analysis & Design

Lecture 1

Agenda

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1. Essential module information
2. Module assessment
3. Module outline
4. What is System Analysis & Design and Why do you need it?
5. Systems Development Life Cycle (SDLC)
6. Major Software Development Models / Categories

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Module Introduction

Essential module information

Essential Module Information

Module leader

- **Said Abduvaliev**
- sabduvaliev@wiut.uz
- Office hours: Fridays 11:00-13:00

Module team

- **Isomiddin Abdunabiev**
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- Office hours: Thursday: 15:00 - 17:00

Module Assessment

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1. Group Coursework – 40%
2. Individual Coursework– 60%

Content outline

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Lectures (theory)

- Information Systems
- Methodologies and frameworks
- Tools
- Techniques
- Object-oriented analysis and design

Tutorials (practice)

- Application of the theory
- Techniques to manage teams effectively
- IT project management
- Apply methodologies in your context

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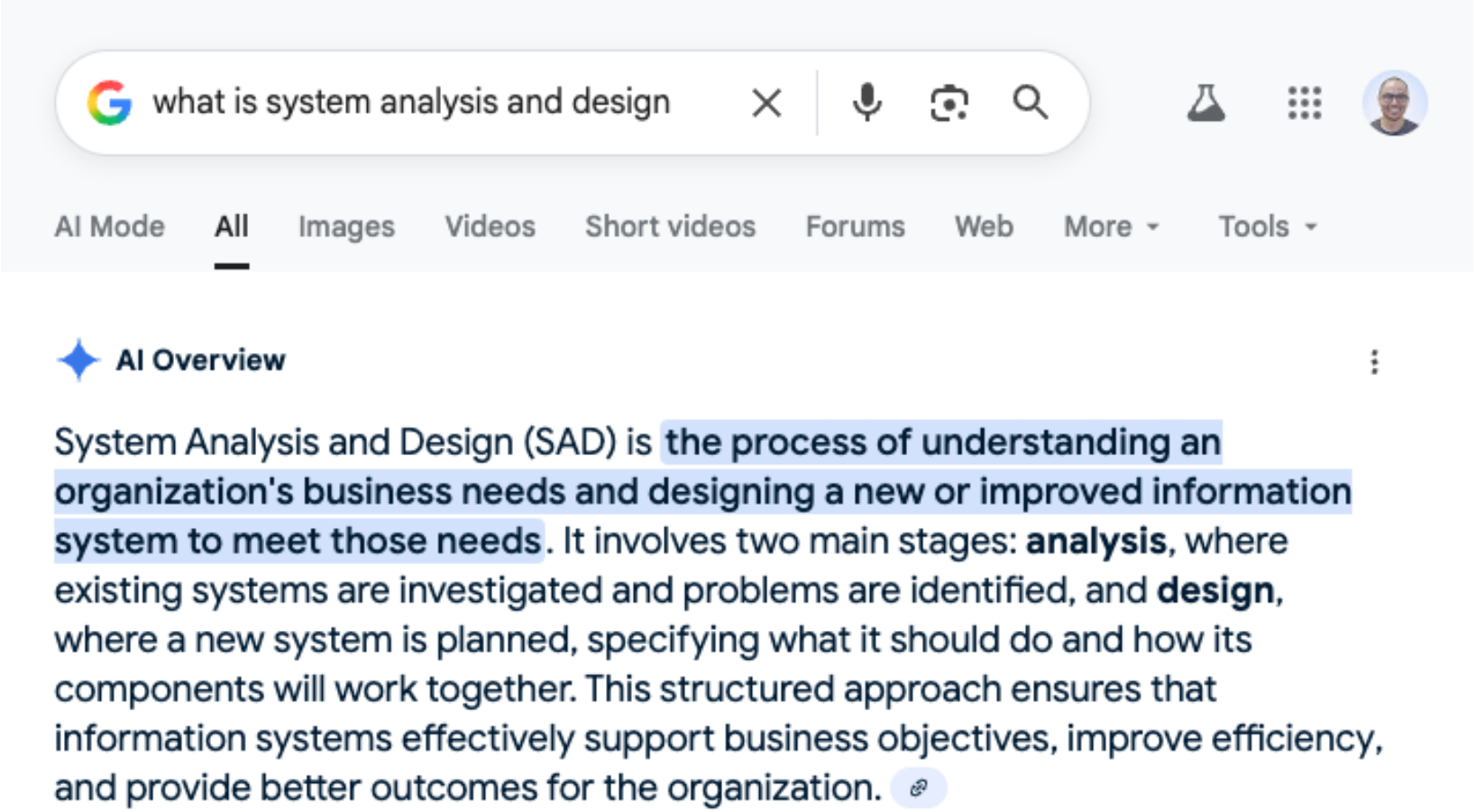
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What is SAD?

What is your guess?

Let’s ask Google

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Why SAD?

Why would you possible need SAD?

Why SAD?

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Teaches:

- Effective project management.
- Developer teams management.
- Effectively document system requirements and solutions/designs.
- Helps with project success.

SAD in AI age

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Analyse - Understand the business problems and issues. Gather requirements and come up with a solution.

This is basically building the essential context for your AI agent to build the right solution.

Design – Standard way to communicate your systems architecture, code, design.

A “language” to communicate system architecture and design to your AI agent, so that it doesn’t generate AI slop.

SAD in AI age

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- AI is replacing junior positions. According to a Stanford's SDE Lab, the entry position employment is dropping19% annually[1].
- However, project and team management roles are not affected by AI.
- SAD teaches you how to effectively manage developer teams and projects.

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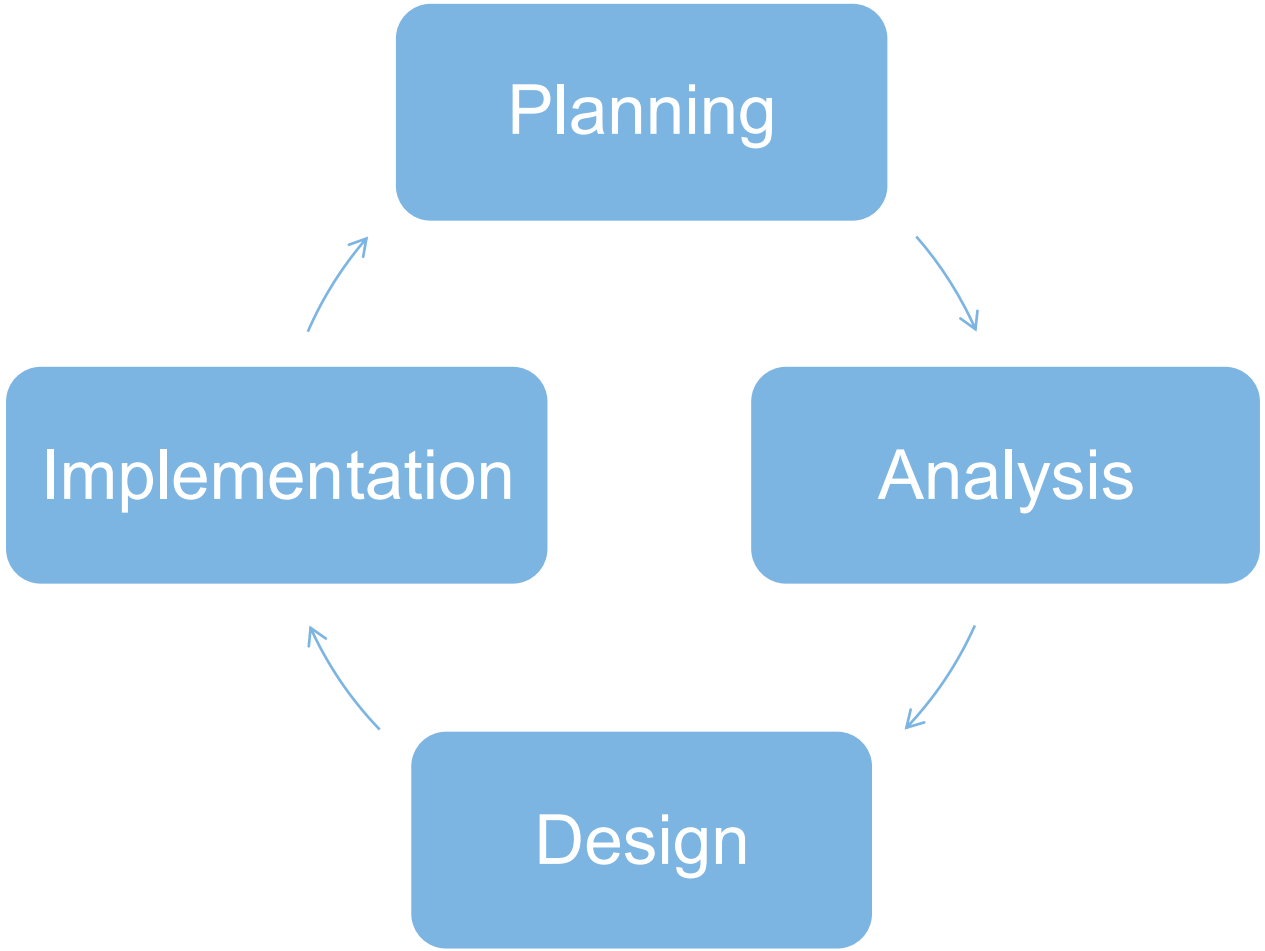
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Systems Development Life Cycle

What is SDLC?

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SDLC: Planning

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- Project initiation
- Feasibility study
- Develop a plan

- **Outcome:** Clearly understand and document why the new IT system is being built.

SDLC: Analysis

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- Decide on analysis strategy
- Carry out requirements gathering
- Propose a system solution / vision

- **Outcome:** Clearly understand what the solution should do.

SDLC: Design

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- Develop a design strategy
- Develop database design
- Propose a system design solution

- **Outcome:** Clearly understand how the solution should be built.

SDLC: Implementation

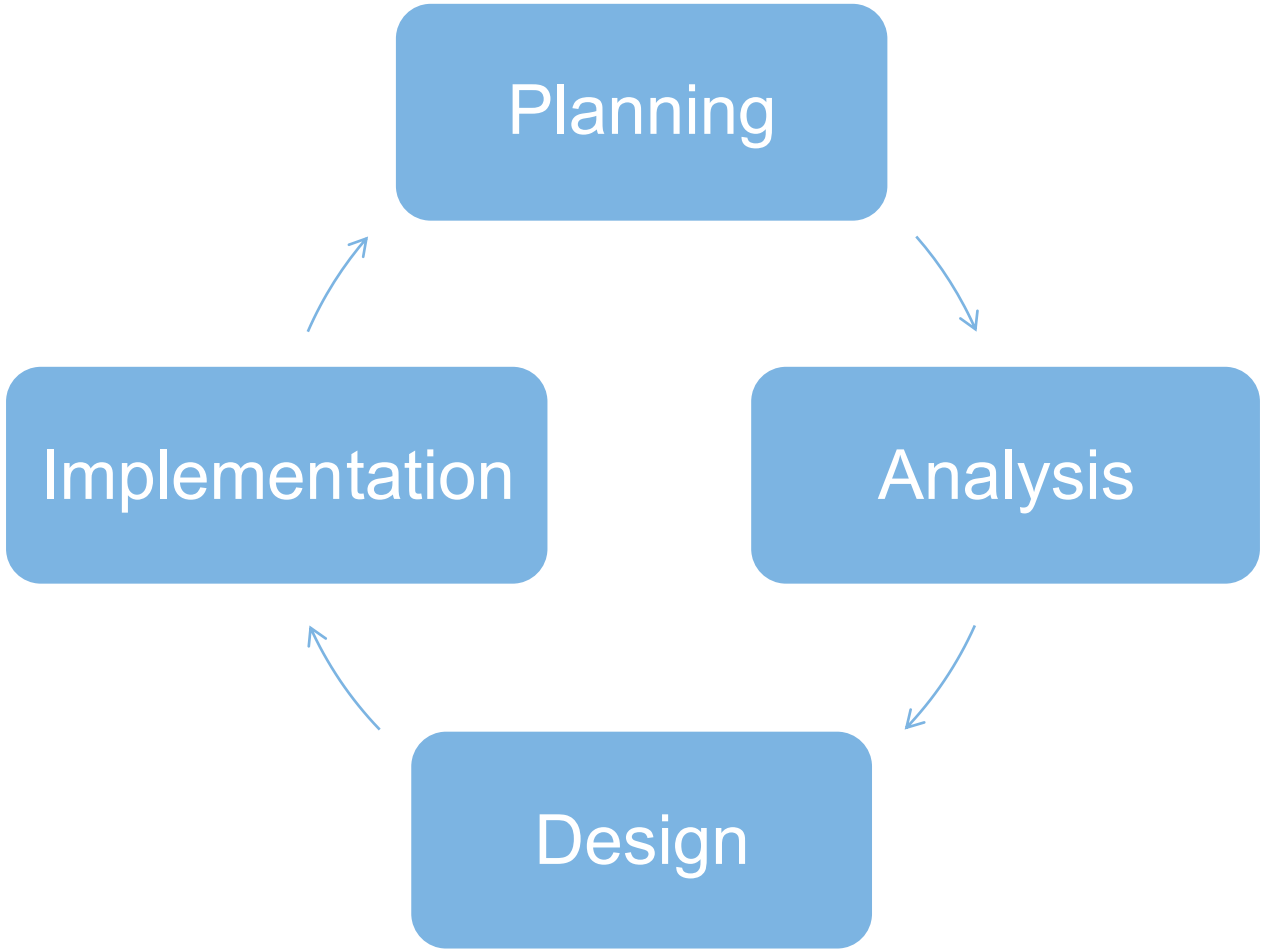
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- Build the system
- Deploy/install the system
- Maintain the system

- **Outcome:** Build and maintain the system.

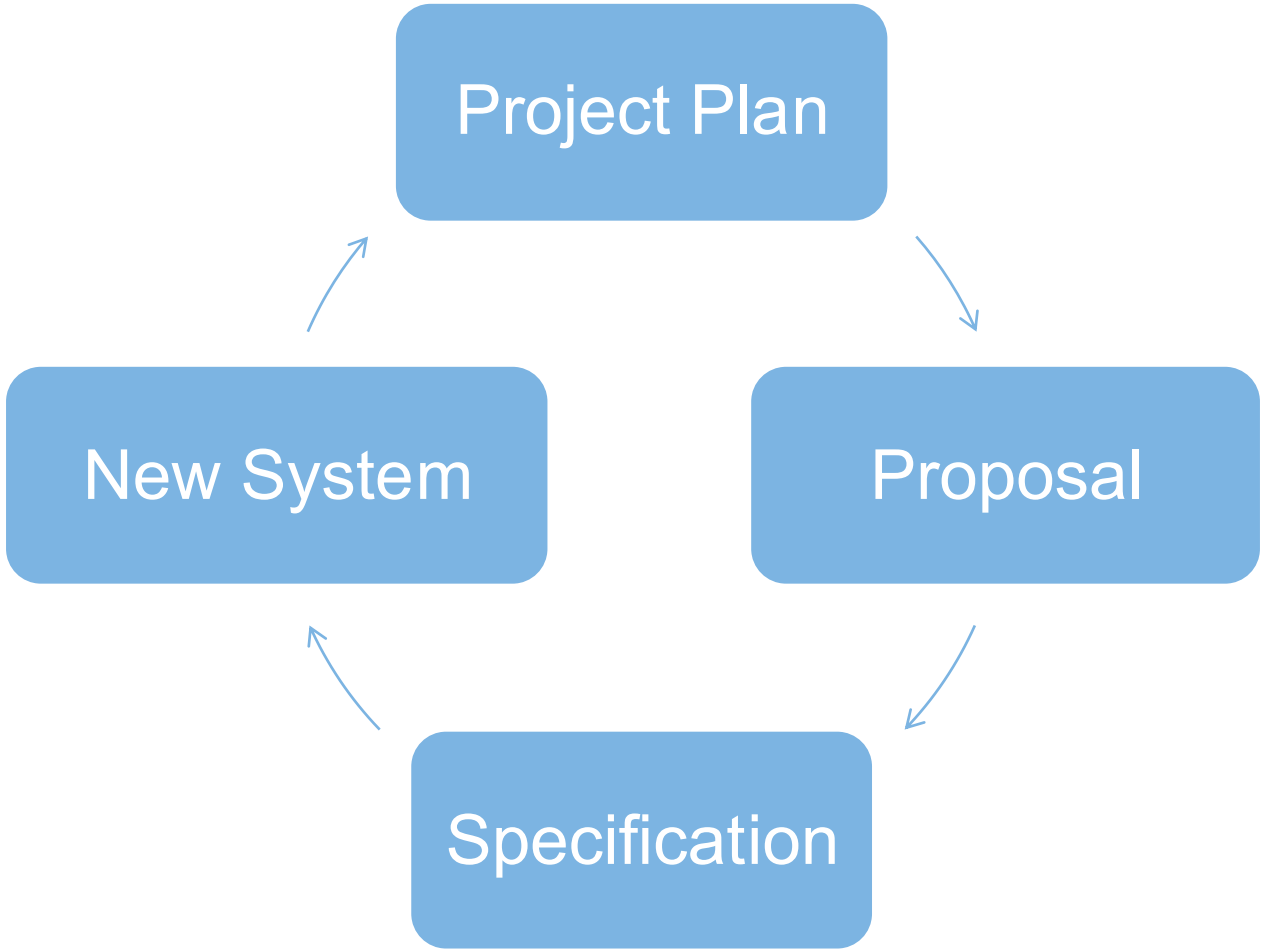
SDLC

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Deliverables

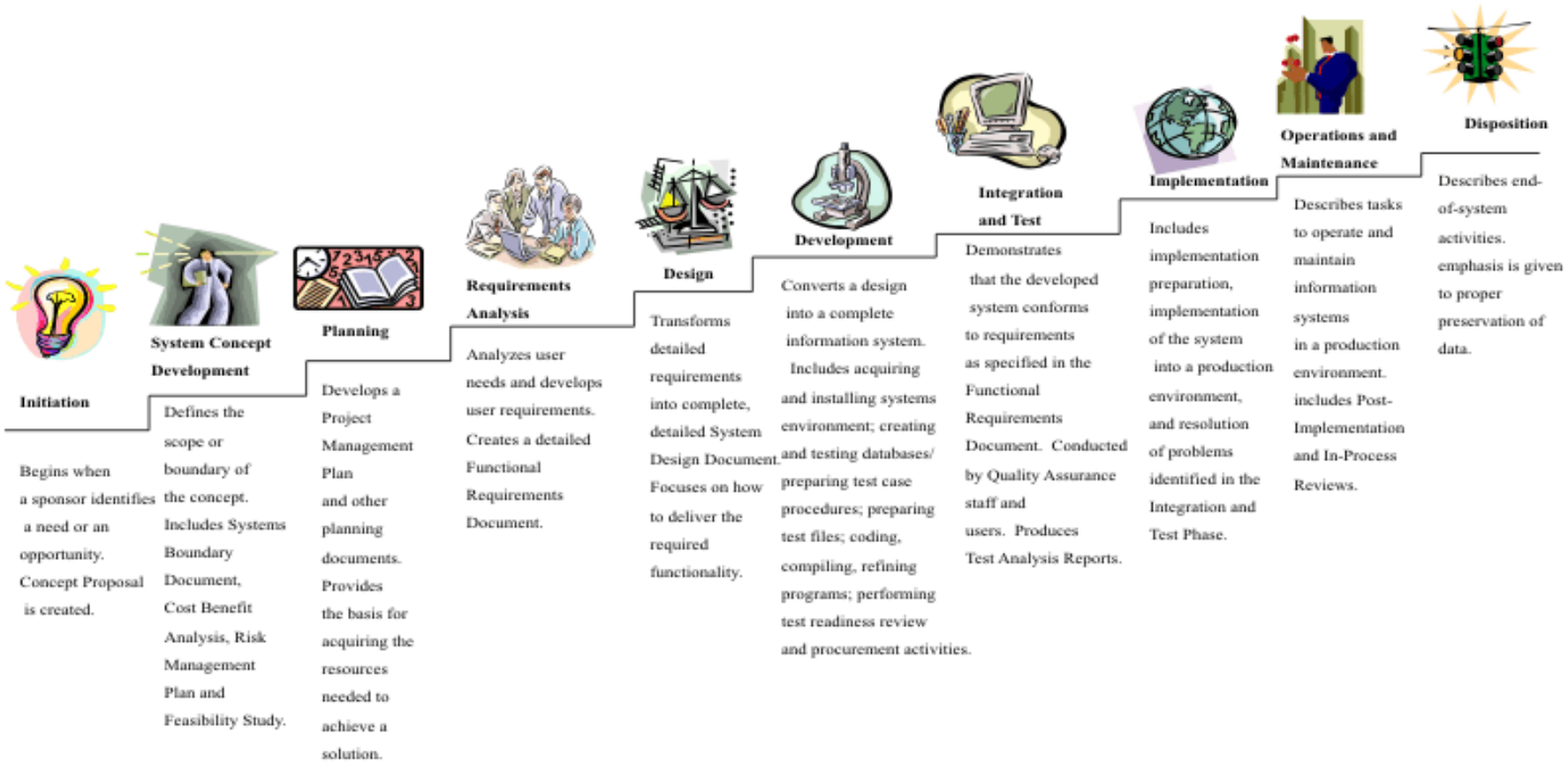
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SDLC

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Systems Development Life Cycle (SDLC)
Life-Cycle Phases



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Systems Development Methodologies

Frameworks to structure your project.

Methodologies are…

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- Concrete implementation of SDLC.
- Most know methodologies:
  - Waterfall
  - Agile (Scrum, XP, etc.)
  - Rapid Application Development (RAD)
  - Prototyping
- Major Software Development Models / Categories
  - Waterfall (structured, top/down)
  - Iterative & Incremental (spiral)
  - Prototyping

Methodologies are…

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There are many Software Development Models with their strengths and weaknesses. But they can be categorized into 3 major categories:

1. **Waterfall** – Structured, top/down approach.
   Examples: Waterfall, Structured Software Development Methodology (SSDM).

2. **Iterative & Incremental** – Iteratively go through important software development "steps" (analysis, design, build, etc.) and incrementally build the system.
   Examples: Spiral, RAD, Agile, etc.

3. **Prototyping** – Build prototypes, get feedback, refine. Once get the final version – build the software and deploy.
   Examples: RAD, Prototyping.

Waterfall

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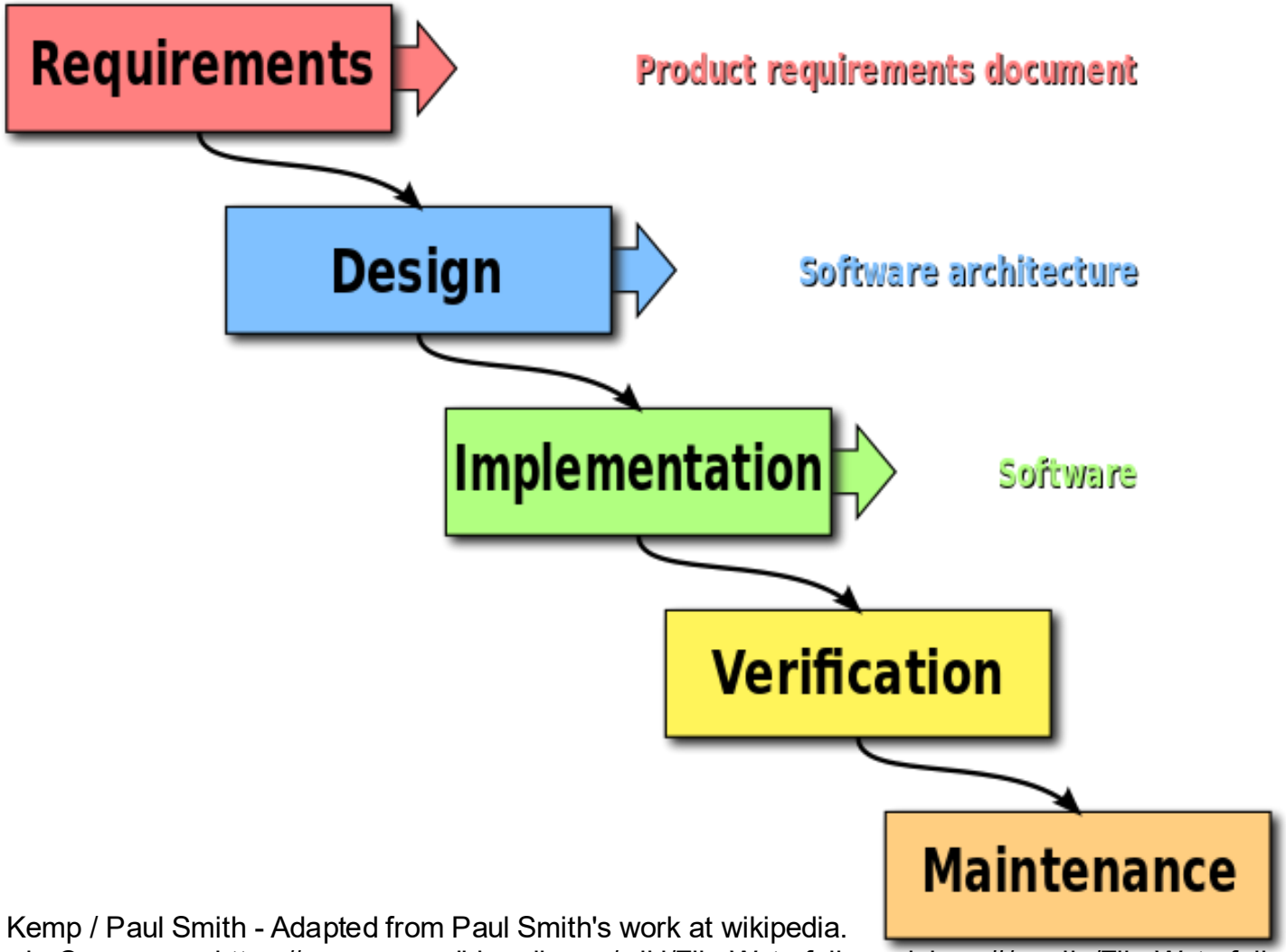
The waterfall model is a sequential design process, used in software development processes, in which progress is seen as flowing steadily downwards (like a waterfall) through the phases of:

1. Conception
2. Initiation
3. Analysis
4. Design
5. Construction
6. Testing
7. Production/implementation
8. Maintenance

Source: https://en.wikipedia.org/wiki/Waterfall_model

Waterfall

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"Waterfall model" by Peter Kemp / Paul Smith - Adapted from Paul Smith's work at wikipedia.
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Iterative & Incremental

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Instead of planning everything upfront, Iterative & Incremental approaches to software development utilize unique features of software.

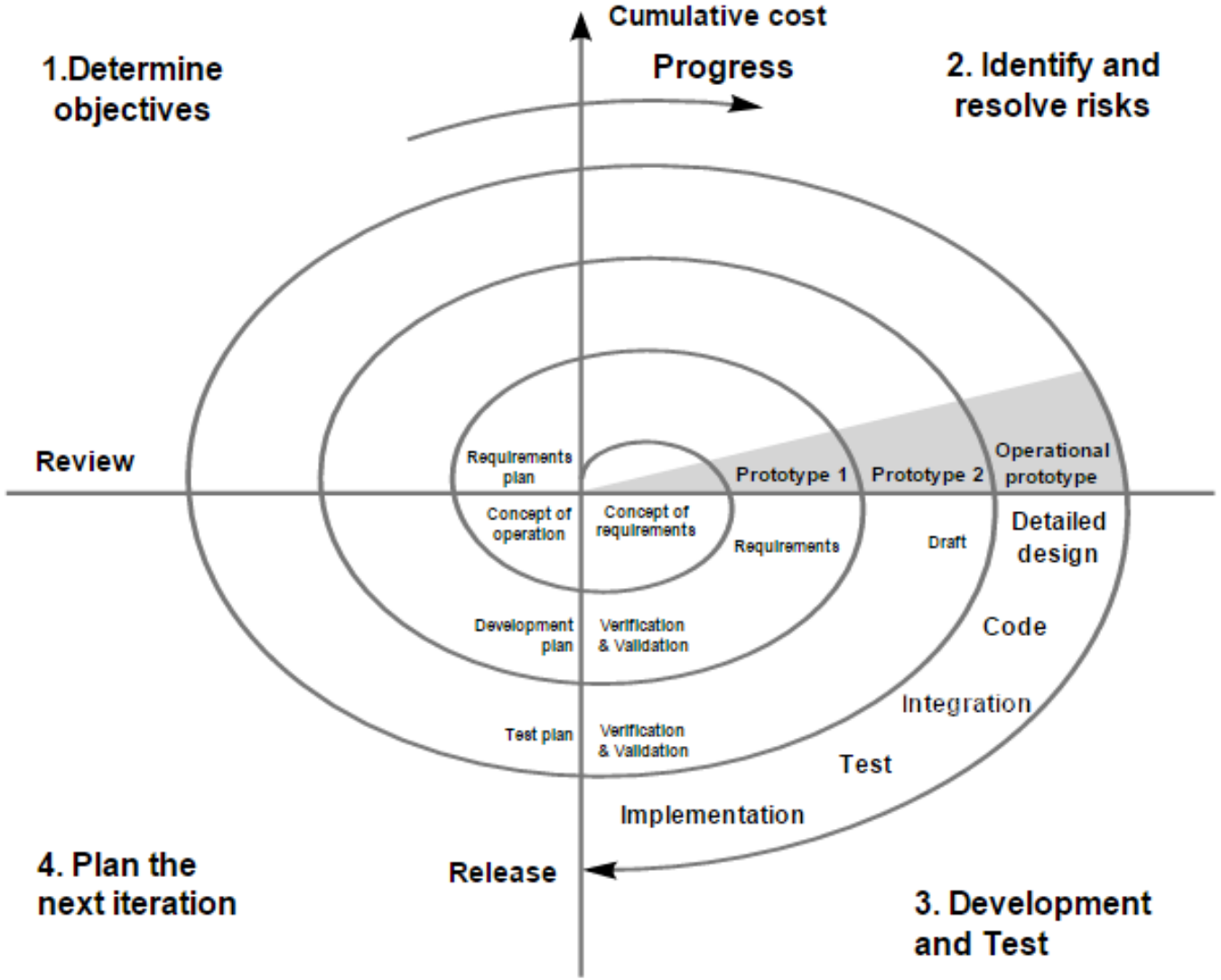
Compared to other “products” in construction or manufacturing, software is easier and cheaper to change at later stages.

However, never forget, that it is **always** more expensive to change any product in later stages of development.

And the approaches and methodologies that we will learn, will help us reduce these risks (related to cost, time, resources, etc.) and help us increase our project success probability.

Iterative & Incremental

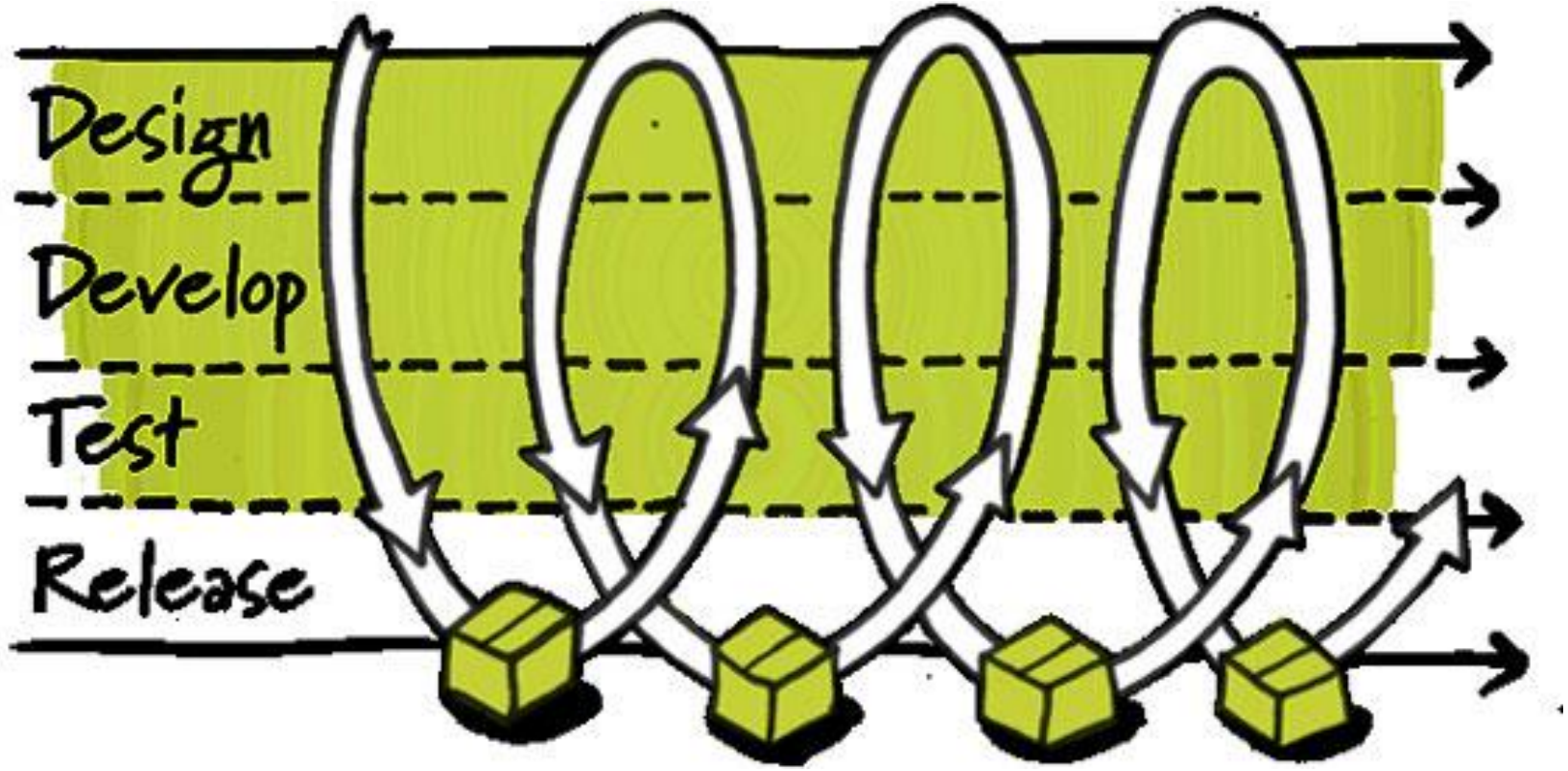
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Iterative & Incremental

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Jason Oh, Essential Resources, Services & Technologies for Your Startup. <https://www.linkedin.com/pulse/essential-resources-services-technologies-your-startup-jason-oh>

Prototyping

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It is hard to collect project requirements correctly right the first time.

Different approaches and methodologies of software development try to deal with this problem differently.

Some spend substantial time and resources to get it the first time (structured).

Some break down the project into smaller parts and iterate through them (iterative).

And others try to solve it by showing the final user prototypes and get their feedback (prototyping).

Anything can be considered a prototype, working or not. For example: drawing, working software, video, HTML version of a desktop app.

Prototyping

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System Develop Methodologies

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We will go though all of these and other methodologies available to you.

We will learn what problems each methodology solves, their strengths and weaknesses in different contexts, advantages and disadvantages.

Basically, when to use specific methodology and when not to!

System Develop Methodologies

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Structural:

- Waterfall
- Structured Software Development Methodology (SSDM)

Iterative and Incremental:

- RAD
- Spiral
- eXtreme Programming
- Scrum
- Etc.

Prototyping:

- Prototyping

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Waterfall Model

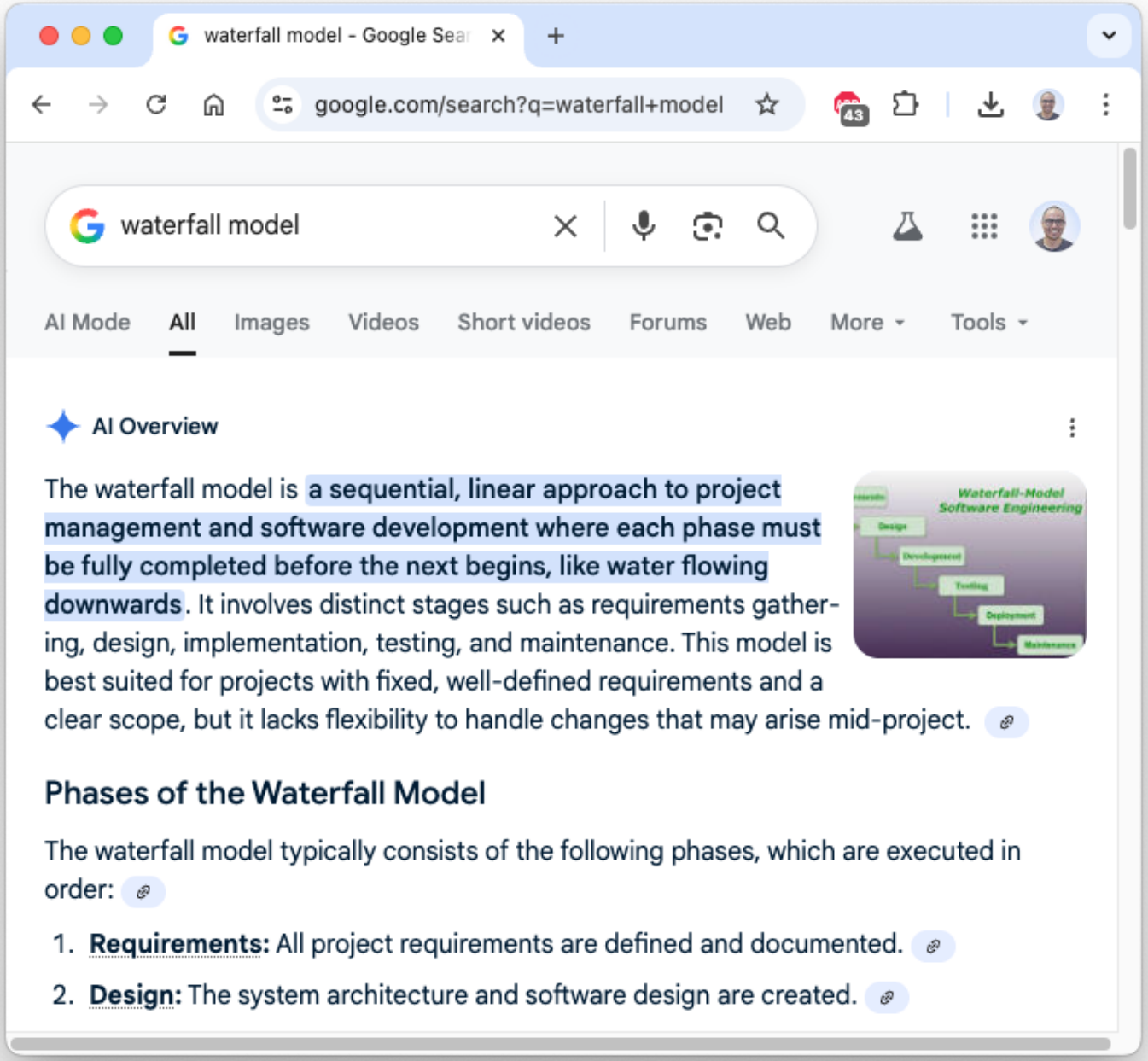
Our first SDLC methodology

Waterfall Model

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The waterfall model is a sequential design process, used in software development processes, in which progress is seen as flowing steadily downwards (like a waterfall) through the phases of conception, initiation, analysis, design, construction, testing, production/implementation and maintenance.

The waterfall development model originates in the manufacturing and construction industries: highly structured physical environments in which after-the-fact changes are prohibitively costly, if not impossible. Since no formal software development methodologies existed at the time, this hardware-oriented model was simply adapted for software development.


waterfall model - Google Sea
google.com/search?q=waterfall+model
waterfall model
AI Mode
All
Images
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AI Overview
The waterfall model is a sequential, linear approach to project management and software development where each phase must be fully completed before the next begins, like water flowing downwards. It involves distinct stages such as requirements gathering, design, implementation, testing, and maintenance. This model is best suited for projects with fixed, well-defined requirements and a clear scope, but it lacks flexibility to handle changes that may arise mid-project.
Waterfall-Model Software Engineering
Phases of the Waterfall Model
The waterfall model typically consists of the following phases, which are executed in order:
1. Requirements: All project requirements are defined and documented.
2. Design: The system architecture and software design are created.

Problems it solves

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Each step has an input and output.

A step's input is an output of the previous step.

What problems does it address?

- Changing project later on is costly. Tens or hundreds of times costly. That's why we need to spend time and resources to get them right at very first step.
- People may leave your team. They may be the carriers of the project related knowledge. That's why you need to document everything in a standard way.
- Gives concrete steps, that are easy to understand. Everyone knows at what phase the project is, and is the main goal at the moment – synergy.

When to use it?

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- Projects that have clearly defined final product.
- Stable and clear requirements i.e. requirements do not change.
- Clearly defined and fixed scope.
- Large scale project. For small projects it is not as effective.

What else do you think?

Advantages & Disadvantages

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From what you have heard so far…

What are the advantages of Waterfall methodology?

We will introduce **disadvantages** of the model, while introducing other methodologies. Usually, a new methodology is trying to solve specific problem in or that was not addressed in some methodology.

The End

Any questions?